

# Exact staged payload execution

## LOCAL EVIDENCE / REMOTE CI NOT YET RUN

**Report ID** 20261001-205927-staged-payload-export

**Prepared UTC** 2026-10-01 20:59:27

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**Parent commit:** 5e73a19f9b910d8bad0f19c8f1f5e56f266e134b

**Candidate SHA-256:** fbddaceaadcaa0691d20212f4c34ff6bce0b4a37f00469bf27295dfaf08da855

Fingerprint excludes only this report's JSON/TEX/PDF; it is not the final commit SHA.

### WHY / problem and value

Contributor validation must run the bytes being committed while preserving unstaged changes.

### WHAT / change scope

Export the complete index into an isolated local Git snapshot under artifacts. Reject unsafe paths, conflicts, unsupported objects and output collisions.

### HOW / implementation rationale

Bind the parent and complete tree, validate entries before writing, preserve modes and internal symlinks, and check identity after export. Prove a passing worktree cannot hide staged exit-seven code.

### WHERE / affected paths

scripts/verification/  
tests/verification/  
package.json; Makefile; makes/verification.mk  
docs/engineering/development.md

### PRESERVED / existing contracts

No stash, reset or checkout in the contributor worktree. Original npm application interfaces and production inputs remain unchanged. Only exact current report paths are excluded from identity.

### RISK / what could go wrong

Snapshots use local Git object sharing and must retain the source object store during checks. Case-colliding paths and escaping symlinks fail rather than execute an ambiguous candidate.

### LIMITATIONS / what is not proved

Six staged-export probes, 13 developer tests and 57 report tests passed on the frozen index with Node 24.21.0/npm 11.19.0. Export does not itself execute application gates or admit a commit.

### ROLLBACK / restore and verify

Revert export tools, tests and aliases together in a reported change. Existing manual staged inspection and application commands remain available.

## VALIDATION / actual local evidence

| Gate             | Command / check           | Outcome | Duration | Evidence                                                          |
|------------------|---------------------------|---------|----------|-------------------------------------------------------------------|
| Staged execution | npm run verification-test | PASS    | 1.79 s   | artifacts/admission/staged-payload-export/verification-tests.json |
| Report tools     | npm run report-test       | PASS    | 6.09 s   | artifacts/admission/staged-payload-export/report-tests.json       |
| Developer tests  | npm test                  | PASS    | 1.31 s   | artifacts/admission/staged-payload-export/command-tests.json      |

**Remote validation at authoring:** NOT\_RUN. Later CI belongs to the immutable commit's check records, not a follow-up report commit.